ORGANOMETALLIC, INORGANIC POLYMERS, BIOINORGANIC CHEMISTRY

The Compound in which a metal atom is bonded directly to a carbon atom of at least one organic group or radical are known as organometallic compounds. It involves organic group, radical various hydrocarbon's and its derivatives, carbonyls, carbides, chelates etc. But do not include carbonates of metals, complex of metabls with organic amines and metallic salt of organic acids. Thus, trimethyl boron, $B(CH_3)_3$, $Al(C_2H_5)$ is organometallic compounds because there is metal (B) to carbon bond. Examples of some organometallic compounds are $(\pi-C_5H_5)_2Fe$, $(C_2H_5)_2Hg$, $(C_2H_5)_4Si$, etc.

Organometallic compounds are of two types.

1. Simple organometallic Compounds

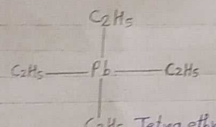
Simple organometallic compounds are those in which only hydrogen atom or hydrocarbon radical attached to the metal atom. These compounds are subdivided into two types.

a. Symmetrical eg. $(C_2H_5)_2Zn$, $(C_2H_5)_4Pb$

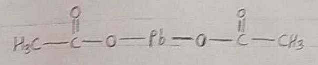
b. Unsymmetrical eg. $C_2H_5HgCH_3$

2. Mixed organometallic compounds

Mixed organometallic compounds are those in which hydrogen or hydrocarbon radical and other group directly attached to the metal atom for example. RMgX, C_2H_5MgBr, $C_4H_9SnCl_2$, etc.



Tetra ethyl lead



Lead acetate

From the above in the first compound the Pb atom is directly attached to carbon atom of ethyl group. Hence it is an organometallic compound. But in the second compound the Pb atom is not directly attached to carbon atom while it through oxygen atom. Hence, it is not an organometallic compound.

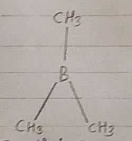
- **CLASSIFICATION OF ORGANOMETALLIC COMPOUNDS**

Organometallic compounds can be classified into three types on the basis of nature of metal-carbon bond.

i) Sigma bonded Compounds

In this type of compounds sigma is formed between metal and carbon. The few examples of this type are- Grignard's reagents (R-Mg-X) like $CH_3-Mg-Br$: Methyl magnesium bromide $Zn(C_2H_5)_4$: Tetra ethylzinc.

Some examples of organometillic compounds are –



Trimethyl Boron

$(C_2H_5)_4Pb$ Tetraethyl Lead

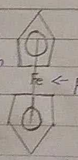
$CH_3-Mg-Br$ methyl magnesium bromide

ii) Pi bonded Compounds:

When in the compounds there is a bonding between metal and the cloud of delocalised electrons of organic molecules. This is a π bonding and hence the compounds are called pi bonded complex.

For example

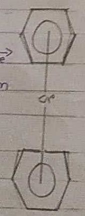
Ferrocene, each cyclopentadiene molecule contributes five electrons to form compound when as in bis benzene chromium, each nzene ring contributes six electrons to form compound.



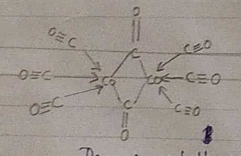
Ferrocene

iii) Multiple bonded Compounds

When in compounds there are multiple bonds formed between metal and carbon atoms then it is called as multiple bonded compounds.



Bis-Benzene Chromium



Dimeric cobalt carbonyl

The difference between organometallic compounds and any other organic compound containing metal atom will be clear from following examples. i) Tetraecthyl lead. ii) lead actale.

- **MODERN CLASSIFICATION OF ORGANOMETALLIC COMPOUNDS**

1. Nomenclature of simple compounds

The simple alkyl or aryl organometallic compound of metals are named by writting the name of the metal after the name of organic group, for example, CH_3Li - methyl lithium, $(C_2H_5)_2Zn$ - Diethyl zinc, C_2H_5MgBr - Ethyl magnesium bromide.

2. Nomenclature of Carbonyle

i) The compounds containing Co ligands are called metal carbonyls. In case, the metal has zero oxidation state, it may be mentioned. example:- $Ni(CO)_4$ - Tetracarbonyl nickel.

ii) If the ligands act as bridges between two metal atom, the Greek letter mu (μ) is written before their names. The Prefix μ is repeated before the name of each kind of bridging ligand. For example :- $[(CO)_3Co(CO)_2Co(CO)_3]$ - Di-μ-Carbonyl bis.

iii) When the metal carbonyls contain metal-metal bonds these may be classified as symmetrical or unsymmetrical. The symmetrical metal carbonyls are named by the use of multiple prefixes. In case of unsymmetrical metal carbonyls, one centre metal atom it's ligands are treated as a ligand on the other central metal atom.

For example :- $[(CO)_4Co-Co(CO)_4]$ symmetrical Bis (tetracarbonyl cobalt) unsymmetrical $[(CO)_4Co-Re(CO)_5]$ Penta carbonyl

3. Nomenclature of σ and π bonded ligands

To distinguish between one carbon bonded ligands and multiple carbon bonded ligands, the notations σ and π are used.

- **Essential & Trace elements in Biological Process**

In spite of carbon, hydrogen, oxygen & nitrogen, the mineral elements present in animal body can be classified into two categories.

- **Classified into Two Categories**

Essential elements & non essential elements. As many as twenty elements including metals & non metals are believed to be in life process of human being.

A) Essential Elements

These are the elements which are absolutely necessary for life process depending upon their absolute amounts in the living system, it is further sub divided into two categories.

1) Micro Elements :- These are the elements which needed for the life process in very small amounts. example :- Cu, Zn, Co, Mn, Mo, etc.

2) Macro Elements :- These are the elements which required for the life process in a large amounts. example :- C, H, O, N, Na, etc.

B) NON Essential Elements

These are the elements which are absolutely not needed for the life process because their function can be carried out by other elements. eg :- Al, B, As, etc.

- **FACTORS FAVOURING THE OXYGEN TRANSPORT PROCESS.**

Following factors favour the oxygen transport process by myoglobin and hemoglobin.

1. Pressure of oxygen

Oxygenation of hemoglobin is favoured at high pressure of oxygen while oxygenation of myoglobin is favoured at low pressure of oxygen. In lungs there is high pressure of oxygen which facilitates absorption of oxygen by hemoglobin to convert it to oxyhemoglobin. One the other hand, in the muscle cell oxygen pressure is low which facilitates easy relase of oxygen from oxyhemoglobin and easy absorption of oxygen by myoglobin.

2. PH of the Medium

The oxygenation and deoxygenation of hemoglobin is PH dependent. At low PH i.e. in acidic medium hemoglobin easily bind oxygen. In muscle cells as there is high concentration of CO_2 the PH is high and this favours easy release of oxygen from oxyhemoglobin to myoglobin.

Topic

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Brijlal Biyani Science College, Amravati.

Department of chemistry

Name :- Aakanksha. K. Balode

Enrollment no :- 18125119

Roll No :- 72884

Class :- Bsc 3rd yr (sem 6) BTZ

Subject :- Organometric, Inorganic Polymers, Bioinorganic chemistry.

Date of submission :- 11-8-2021

Signature of student :- Aanksha